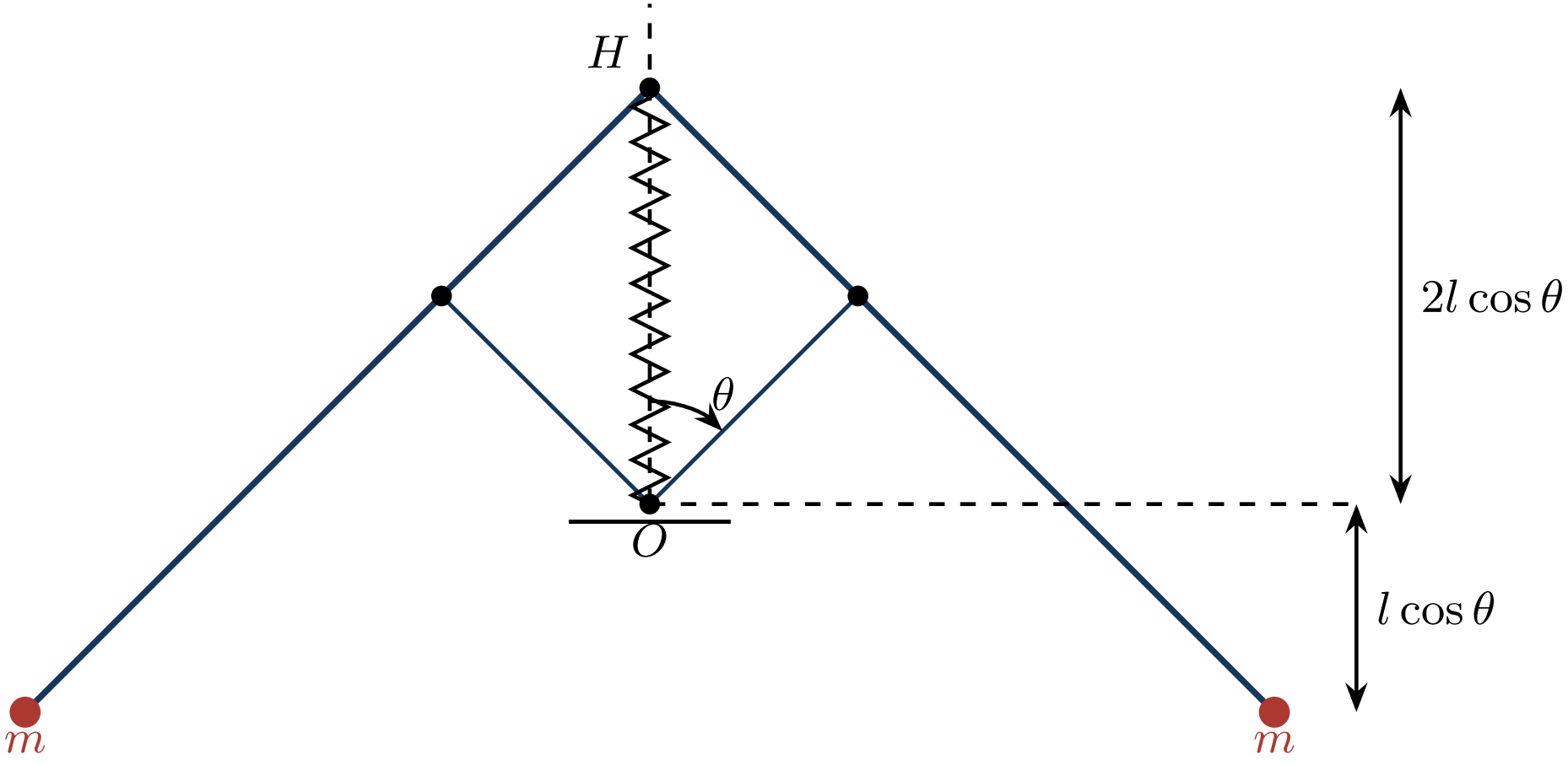

端点：$x = \pm 3l\sin\theta$，$z = -l\cos\theta$；弹簧长度 $L_{\mathrm{s}} = 2l\cos\theta$。